MIL-V-45647(Ord)
10 May 1961
SUPERSEDING
FA(FC)-PD-521
12 August 1959

# MILITARY SPECIFICATION

# VERTICAL ANGLE UNIT: 8644072

## 1. SCOPE

1.1 This specification covers one type of vertical angle unit used to convert a mechanical input of vertical angle to an electrical value in the Gun Direction Computers: M15 and M15C which are part of Field Artillery Fire Control Systems: M35 and M35C.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of |
| MIL-F-14619 | Fire control Parts, Equipment and Tools; Packaging of |
| MIL-I-45208 | Inspection Requirements; General Specification for |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts |

FSC 1220

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

DRAWINGS

Ordnance Corps

| | |
|---|---|
| D8643845 | Electrical Schematic, Vertical Angle Unit |
| F8644072 | Vertical Angle Unit |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one vertical angle unit, representative of production, shall be made available to the Government to determine that the vertical angle unit meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The vertical angle unit, shall be manufactured in accordance with Drawing F8644072, "Vertical Angle Unit", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential.- The design and construction of the vertical angle unit shall insure that all external parts are at ground potential at all times.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

3.5 Environmental.

3.5.1 Vibration.- The vertical angle unit shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.03 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the vertical angle unit shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperature.- The vertical angle unit shall meet requirements of 3.6 through 3.10 when thermally stabilized at ambient temperatures of plus 150 and minus 40 degrees Fahrenheit.

3.5.3 Storage temperatures.- The vertical angle unit shall show no evidence of physical failure and shall meet the requirements of 3.6 through 3.10 at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 and minus 65 degrees Fahrenheit.

3.6 Counter.- The counter shall be capable of rotating to at least 300 mils in either direction from zero.

3.7 Backlash.- The backlash of the counter gear train shall be no greater than 0.5 mil as determined from the reading of the counter.

3.8 Circuitry.- Circuitry shall be in accordance with Electrical Schematic, Drawing D8643845.

3.9 Electrical and mechanical output.- Output voltages shall be within the tolerances specified in Table I as a result of clockwise and counterclockwise rotation of the input knob. The electrical output signal shall be measured when loaded with 100,000 ohms plus or minus 0.5 percent from wiper to center-tap of the variable resistor. The reference voltage applied across the variable resistor shall be plus and minus 15 volts, 400 cps as referred to a common neutral.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

TABLE I

| Counter Readings (Mils) | Output Voltages (Volts) | Tolerance (Millivolts) |
|---|---|---|
| -300 | 4.529 | ± .025 |
| -240 | 3.609 | ± .020 |
| -180 | 2.688 | ± .015 |
| -120 | 1.784 | ± .010 |
| - 60 | 0.900 | ± .010 |
| 0 | 0.000 | ± .010 |
| + 60 | 0.900 | ± .010 |
| + 120 | 1.784 | ± .010 |
| + 180 | 2.688 | ± .015 |
| + 240 | 3.609 | ± .020 |
| + 300 | 4.529 | ± .025 |

3.10 Torque.- The torque required for rotation of the input knob shall be between 10 and 20 ounce-inches through the full range of travel at the temperatures specified in 3.5.2. Slippage of the clutch shall occur between 2 and 3 pound-inches.

3.11 Illumination of counter.- The lamp for the window of the counter shall be illuminated when energized from a 28 volt dc source and shall provide enough illumination to make all meter readings legible throughout the entire range.

3.12 Alinement and calibration.- Alinement and calibration of all observer vertical angle units shall be performed using the same power source to insure uniformity of product. The simulated reference voltages supplied shall be from a power source maintained as a standard.

3.13 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order
(b) Contractor's quality standards of acceptability
(c) Procedure for continually evaluating product quality
(d) Procedure for continually reporting evidence of actual product quality
(e) Measures for preventing any significant departure of product quality from standards

4.1.1 The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification shall consist of: (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Contractor's inspection system.- Contractor's inspection and Government verification of contractor's inspection shall be in accordance with Specification MIL-I-45208.

4.4 Production model.- Unless otherwise specified in the contract, one vertical angle unit shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.6 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the vertical angle unit. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.6.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table II.

TABLE II - SAMPLING PLAN

| Lot Size Inclusive | Sample Size | Accept Lot | Reject Lot |
|---|---|---|---|
| 0 - 25 | Use 100% Inspection | | |
| 26 - 35 | 20 | 0 | 1 |
| 36 - 50 | 25 | 0 | 1 |
| 51 - 150 | 30 | 0 | 1 |
| 151 - 300 | 30 | 1 | 2 |
| 301 - over | 60 | 2 | 3 |

4.6.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

4.7 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.8 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.8.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.2 Accuracy requirements for special testing devices.

4.8.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion as specified. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in the specified positions.

4.8.2.2 Devices to test torque.

4.8.2.2.1 Input knob.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to twenty-five ounce-inches, the accuracy of which shall be within plus or minus one-quarter ounce-inches throughout the entire zone of excursion. The device shall be equipped with an adapter to accommodate the input knob.

4.8.2.2.2 Clutch.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to five-pound-inches, the accuracy of which shall be within plus or minus one-quarter pound-inches throughout the entire zone of excursion. The device shall accommodate the same adapter used for testing the input knob torque requirements.

4.9 Tests.

4.9.1 Classification of defects.- The classification of defects in Tables III and IV shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.9.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

| MAJOR: AQL 4.0% | Req't. Par. | QA. Par. | Test Method |
|---|---|---|---|
| 101. Post vibration | 3.5.1 | 4.9.2 | Device, Special Testing (DST)-Visual |
| 102. Operating temperatures | 3.5.2 | 4.9.3 | DST-Visual |

MINOR: NONE DEFINED

TABLE IV - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

| MAJOR: AQL 1.5% | Req't. Par. | QA. Par. | Test Method |
|---|---|---|---|
| 101. Counter | 3.6 | 4.9.5 | Visual |
| 102. Backlash | 3.7 | 4.9.6 | Standard Meas. Equipment (SME) |
| 103. Circuitry | 3.8.1 | 4.9.7 | Visual |
| 104. Electrical and mechanical output | 3.9 | 4.9.8 | SME |
| 105. Torque | 3.10 | 4.9.9 | Device, Special Testing (DST) |
| 106. Illumination | 3.11 | 4.9.10 | SME |
| 107. Identification marking | 3.13 | 4.9.11 | Visual |

MINOR: NONE DEFINED



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

4.9.2 Vibration.- This test shall be conducted in accordance with Method 201 of Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements, prior to or during vibration, are not required for final acceptance. Subsequent to vibration, the vertical angle unit shall be subjected to a visual-tactile examination for damage. The unit shall then be subjected to all the tests in this specification.

4.9.3 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the vertical angle unit is stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.9.5 to 4.9.10 inclusive and shall meet the requirements specified.

4.9.4 Storage temperatures.- This test shall be conducted in accordance with Procedure II, condition "B" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the vertical angle unit shall be subjected to tests specified in 4.9.5 to 4.9.10 inclusive and shall meet the requirements specified. This test shall be performed on the production model only.

4.9.5 Counter.- A visual check shall be made to determine if the specified number of mils rotation can be made.

4.9.6 Backlash.- Energize the unit and advance the counter to some reading; record the output signal for that reading. Continue rotation of the counter in the same direction for one revolution, then reverse rotation and return until the output signal repeats the recorded reading. The counter shall have repeated its reading within the specified tolerance.

4.9.7 Circuitry.- A visual check shall be made to be assured that the circuits conform to the prescribed electrical schematic.

4.9.8 Electrical and mechanical output.- Energize the unit and apply the specified voltages and resistance to the applicable points. The counter readings are then set, by means of the input knob, to the mil settings of Table I. The output voltages shall be measured by means of standard measuring equipment and shall be within the millivolt tolerance.

4.9.9 Torque.- This test shall be performed with the aid of special testing devices, affixed to the input knob, that shall be capable of applying torque to the knob within the ranges specified.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

The torque required for rotation of the input knob and for clutch slippage shall be within the values specified.

4.9.10 Illumination.- This test shall be performed with the lamp energized as specified. All graduations of the counter shall be legible when observed, with the unaided eye, in total darkness.

4.9.11 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The vertical angle unit shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619/P8538. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 Intended use.- The vertical angle unit shall be capable of receiving and recording a mechanical input of observer vertical angle, designated in mils measured as positive above the horizon and negative below the horizon, and converting this input to an electrical output value. This output value is equivalent to the tangent of observer vertical angle, which is proportional to the product of observer distance and observer vertical angle. The vertical angle unit is a functional plug-in subassembly of the Gun Direction Computers: M15 and M15C.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels or preservation packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment (SME).



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.

6.4.2 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.3 Tactile.- Skillful manipulation or sensing.

6.4.4 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodian:

Army-Ord

Preparing Activity:

Army-Ordnance



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:38Z
Check the source to verify that this is the current version before use.